Strand 1: Number

Sub-strand 4: Ratios and Proportions

Lesson 1: Ratio, Proportion, Rates, Wages, and Financial Arithmetic

1. Curriculum Standard and Overview

- Strand: 1 — Number
- Sub-strand: 4 — Number: Ratios and Proportion
- Content Standard: B9.1.4.1
- Indicators:
- B9.1.4.1.1 — Represent proportional relationships by equations.
- B9.1.4.1.2 — Use proportional relationships to solve multi-step ratio and percent problems (money, wages, income).
- B9.1.4.1.3 — Use knowledge of rates and proportional reasoning (speed, work, pumps).
- Mock-paper links: P2 Q1(b) (income over 5 years), P2 Q3(a) (wages + overtime), P2 Q5(c) (pump rate), P1 Q6 (men and days), P1 Q23 (speed).

2. Preamble — Proportion Is Everywhere

If 1 kg of rice costs GH₵ 12, then 3 kg costs GH₵ 36. The cost is proportional to the quantity — double the rice, double the cost.

Three related ideas:

1. Ratio — comparing two quantities: 3 : 5.
2. Proportion — two ratios that are equal: ⅗ = 6/10.
3. Rate — a comparison with different units: 20 gallons per minute, 80 km/h.

3. Ratio

3.1 Simplifying a ratio

Divide both parts by their HCF. 18 : 24 = 3 : 4 (dividing by 6).

3.2 Sharing in a given ratio

Share GH₵ 720 in the ratio 2 : 3.

1. Total parts = 2 + 3 = 5.
2. One part = 720 ÷ 5 = 144.
3. Shares: 2 × 144 = 288; 3 × 144 = 432.

Check: 288 + 432 = 720 — correct.

3.3 Ratio to fraction

2 : 3 means ⅖ and ⅗ of the total.

4. Direct Proportion

If y is directly proportional to x (written y propto x), then y = kx for some constant k.

Find k first, then use it.

Example: y propto x and y = 15 when x = 5. 15 = k × 5 → k = 3. So y = 3x. When x = 8, y = 24.

Money: Total cost T = p × q (unit price × quantity).

5. Inverse Proportion

If y is inversely proportional to x (y propto 1/x), then xy = k (constant).

Classic: men and days. More men → fewer days (same work).

$$\text{men}_1 \times \text{days}_1 = \text{men}_2 \times \text{days}_2$$

Mock P1 Q6: 9 men finish in 51 days. How long for 17 men? 9 × 51 = 17 × d → 459 = 17d → d = 27 days. Answer: C.

6. Rates: Speed, Wages, Pumps

6.1 Speed

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Mock P1 Q23: 154 km in 1 ¾ hours = 1.75 h. Speed = 154 ÷ 1.75 = 88 km/h. Answer: B.

6.2 Unit-rate method (pumps, taps, printers)

Mock P2 Q5(c): 20 gallons/minute for half an hour (30 min). 20 × 30 = 600 gallons.

6.3 Wages and overtime

Mock P2 Q3(a): Mark earns GH₵ 8.25/hour for 8 hours, plus overtime at 1 ½ times the regular rate for 4 hours.

- Regular pay: 8.25 × 8 = 66.00
- Overtime rate: 1.5 × 8.25 = 12.375
- Overtime pay: 12.375 × 4 = 49.50
- Total: 66.00 + 49.50 = GH₵ 115.50

6.4 Income over a period

Mock P2 Q1(b): Fred earns GH₵ 2500/month after tax. Eben earns three times as much.

- Eben: 3 × 2500 = 7500 per month.
- Combined monthly: 2500 + 7500 = 10 000.
- Five years = 5 × 12 = 60 months.
- Total: 10 000 × 60 = GH₵ 600 000.

7. Twenty Fully Worked Examples

Example 1 — Simplify ratio

18 : 24 = 3 : 4.

Example 2 — Share money

Share GH₵ 720 in ratio 2 : 3 → GH₵ 288 and GH₵ 432.

Example 3 — Ratio as fractions
2 : 3 → ⅖ and ⅗ of the total.

Example 4 — Direct proportion
y propto x, y = 15 when x = 5. Find y when x = 8. k = 3, so y = 24.

Example 5 — Constant of proportionality (Mock P1 Q22 style)
Line through (2, 6) and (4, 12). k = y/x = 6/2 = 3 (also 12/4 = 3).
Answer: C — 3.

Example 6 — Inverse proportion (Mock P1 Q6)
9 men, 51 days → 17 men: 9 × 51 / 17 = 27 days. C.

Example 7 — Speed (Mock P1 Q23)
154 ÷ 1.75 = 88 km/h. B.

Example 8 — Pump rate (Mock P2 Q5c)
20 × 30 = 600 gallons.

Example 9 — Wages (Mock P2 Q3a)
8.25 × 8 + 1.5 × 8.25 × 4 = 66 + 49.50 = 115.50.

Example 10 — Income over years (Mock P2 Q1b)
(2500 + 7500) × 60 = 600 000 cedis.

Example 11 — Unit price
5 pens cost GH₵ 17.50. Cost of 8 pens: 17.50 ÷ 5 = 3.50 each; 8 × 3.50 = 28.00.

Example 12 — Map scale preview
1 cm represents 4 m. 7.5 cm represents 30 m. (Full scale work in Lesson 10.)

Example 13 — Percent increase
A phone costs GH₵ 400. Price rises by 15%. New price = 400 × 1.15 = 460.

Example 14 — Percent decrease
A GH₵ 200 item is sold at 25% discount. Discount = 50; sale price = 150.

Example 15 — Recipe scaling
A recipe for 4 people uses 300 g rice. For 10 people: 300 × 10/4 = 750 g.

Example 16 — Combined work rate
Kofi paints a room in 6 hours, Ama in 3 hours. Together: ⅙ + ⅓ = ½ of the room per hour → 2 hours.

Example 17 — Fuel consumption
A car uses 8 litres per 100 km. For 350 km: 8 × 3.5 = 28 litres.

Example 18 — Exchange / conversion

GH₵ 1 = 0.08 units of currency X. GH₵ 2500 = 200 X.

Example 19 — Average speed round trip

Go 120 km at 40 km/h (3 h), return 120 km at 60 km/h (2 h). Average speed = 240 ÷ 5 = 48 km/h (total distance ÷ total time).

Example 20 — Ratio with three terms

Share GH₵ 900 in the ratio 2 : 3 : 4. Parts = 9; one part = 100; shares 200, 300, 400.

8. Comprehensive Exercise Sets (40 Questions)

All 5 authentic ratio/proportion/wages/rate mock items are included. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Ratio basics

1. Simplify 18 : 24.
2. Share GH₵ 720 in the ratio 2 : 3.
3. Write the ratio 2 : 3 as two fractions of the total.
4. Share GH₵ 900 in the ratio 2 : 3 : 4.
5. Spiral review: Order ⅜, ½, 5/16.

Exercise 2 — Direct proportion

1. y propto x and y = 15 when x = 5. Find y when x = 8.
2. [Mock 2026 P1 Q22] A line passes through (2, 6) and (4, 12). Find the constant of proportionality.
   A. 1 B. 2 C. 3 D. 6
3. If 5 pens cost GH₵ 17.50, find the cost of 8 pens.
4. A recipe for 4 people uses 300 g rice. How much for 10 people?
5. Spiral review: Evaluate ⅔ × ¼ + ⅖ ÷ 14/15.

Exercise 3 — Inverse proportion (includes mock)

1. [Mock 2026 P1 Q6] 9 men complete a piece of work in 51 days. In how many days will 17 men of the same strength finish the same work?
   A. 29 days B. 31 days C. 27 days D. 37 days
2. If 12 taps fill a tank in 4 hours, how long do 8 taps take?
3. If 6 workers build a wall in 10 days, how many workers are needed to build it in 5 days?
4. Spiral review: Simplify 5^(6) ÷ 5^(2).
5. Spiral review: Which is not equivalent to 3^(6) — 18 or 729?

Exercise 4 — Speed, distance, time (includes mock)

1. [Mock 2026 P1 Q23] A van travels 154 km in 1 ¾ hours. Find its speed in km/h.
   A. 77 km/h B. 88 km/h C. 100 km/h D. 269 km/h
2. A car travels 120 km at 40 km/h and returns 120 km at 60 km/h. Find the average speed for the whole journey.
3. A car uses 8 litres per 100 km. How many litres for 350 km?
4. Spiral review: Amina had 2 ½ litres and drank ⅓ litres. How much remains?
5. Spiral review: Round 12.473 to 2 d.p.

Exercise 5 — Wages and income (includes mock)

1. [Mock 2026 P2 Q3(a)] Mark earns GH₵ 8.25 per hour working for 8 hours. He also earns overtime pay at 1 ½ times his regular rate for 4 hours. How much does he earn altogether in a day?
2. [Mock 2026 P2 Q1(b)] Fred earns GH₵ 2500 a month after tax and Eben earns three times as much. How much is their total income after five years if there are no increases?
3. A worker earns GH₵ 45 per hour for 40 hours a week. What is the weekly wage?
4. The same worker gets a 10% rise. What is the new hourly rate?
5. Spiral review: 112 flies at 0.1 kg each weigh how much?

Exercise 6 — Rates of work (includes mock)

1. [Mock 2026 P2 Q5(c)] A water pump delivers 20 gallons of water per minute. If Kofi can only afford half an hour of pumping, how many gallons will be pumped?
2. Kofi paints a room in 6 hours, Ama in 3 hours. How long working together?
3. A printer prints 12 pages per minute. How many pages in 45 minutes?
4. Spiral review: (0.07 × 0.02) ÷ 14.
5. Spiral review: Classify −3 under $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$.

Exercise 7 — Money, discounts, percentages

1. A phone costs GH₵ 400. Price rises by 15%. Find the new price.
2. A GH₵ 200 item is sold at 25% discount. Find the sale price.
3. A shirt marked GH₵ 80 is sold at a 15% discount. Find the sale price.
4. Find 35% of GH₵ 240.
5. Spiral review: Write 47 300 in standard form.

Exercise 8 — Mixed proportion challenge

1. Two numbers are in the ratio 3 : 5 and their sum is 96. Find the numbers.
2. A map has scale 1 : 50 000. What distance does 4 cm represent in km?
3. If y = kx and y = 21 when x = 7, find y when x = 10.
4. Spiral review: Solve (−3)^(x-2) = −243.
5. Spiral review: In a market of 36 fishmongers, 28 sell mackerel, 18 sell tilapia, 13 sell both. How many sell neither?

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. 3 : 4.
2. One part = 144; shares GH₵ 288 and GH₵ 432.
3. ⅖ and ⅗.
4. Parts = 9; one part = 100; shares 200, 300, 400.
5. 5/16 < ⅜ < ½.

Solutions to Exercise 2

1. k = 3; y = 24.
2. k = 6/2 = 3 → C.
3. 3.50 × 8 = GH₵ 28.00.
4. 300 × 10/4 = 750 g.
5. mathbf{25/42}.

Solutions to Exercise 3

1. 9 × 51 = 17d → d = 27 days → C.

2. 12 × 4 = 8 × t → t = 6 hours.
3. 6 × 10 = w × 5 → w = 12 workers.
4. 5^(4) = 625.
5. 18 (option B).

Solutions to Exercise 4

1. 154 ÷ 1.75 = 88 km/h → B.
2. Total distance = 240; total time = 5 h; average = 48 km/h.
3. 8 × 3.5 = 28 litres.
4. 2 ⅙ litres.
5. 12.47.

Solutions to Exercise 5

1. 66 + 49.50 = GH₵ 115.50.
2. (2500 + 7500) × 60 = GH₵ 600 000.
3. 45 × 40 = GH₵ 1800.
4. 45 × 1.10 = GH₵ 49.50.
5. 11.2 kg.

Solutions to Exercise 6

1. 20 × 30 = 600 gallons.
2. ⅙ + ⅓ = ½ → 2 hours.
3. 12 × 45 = 540 pages.
4. 0.0001.
5. ℤ, ℚ.

Solutions to Exercise 7

1. 400 × 1.15 = GH₵ 460.
2. 200 × 0.75 = GH₵ 150.
3. 80 × 0.85 = GH₵ 68.
4. 0.35 × 240 = GH₵ 84.
5. 4.73 × 10^(4).

Solutions to Exercise 8

1. Parts = 8; one part = 12; numbers 36 and 60.
2. 4 × 50 000 = 200 000 cm = 2 km.
3. k = 3; y = 30.
4. x - 2 = 5 → x = 7.
5. 3.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P2 Q1(b)	Fred 2500, Eben 3×, 5-year total	GH₵ 600 000
P2 Q3(a)	Mark wages + overtime	GH₵ 115.50
P2 Q5(c)	Pump 20 gal/min for 30 min	600 gallons
P1 Q6	9 men 51 days → 17 men	C — 27 days
P1 Q22	Constant of proportionality (2,6)(4,12)	C — 3
P1 Q23	154 km in 1¾ h	B — 88 km/h

11. Lesson Summary

Skill	Formula / Rule
Share in ratio	total ÷ parts, then multiply
Direct proportion	$y = kx$
Inverse proportion	$xy = k$ (men × days)
Speed	distance ÷ time
Wages	rate × hours (+ overtime)
Unit rate problems	rate × time = total

Lesson prepared under `Grade9-Maths/Rules.md` — plain English, British notation, full workings required.